

Median for $N(\mu, \sigma^2) \rightarrow \mu$

$$P(X \leq \text{med}) = \frac{1}{2}$$

$$X \sim N(\mu, \sigma^2)$$

$$P\left(\frac{X-\mu}{\sigma} \leq \frac{\text{med}-\mu}{\sigma}\right) = \frac{1}{2}$$

$$Z = \frac{X-\mu}{\sigma}$$

$$Z \sim N(0, 1)$$

$$P\left(Z \leq \frac{\text{med}-\mu}{\sigma}\right) = \frac{1}{2}$$

$$\Phi\left(\frac{\text{med}-\mu}{\sigma}\right) = \frac{1}{2}$$

$$\Phi(0) = \frac{1}{2}$$

$$\frac{\text{med}-\mu}{\sigma} = 0$$

$$\text{med} = \mu$$

log-normal distribution

$$Y = \log X \sim N(\mu, \sigma^2)$$

X following a log normal distribution with parameters μ , and σ^2 .

$$X = e^Y$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$$

$$-\infty < y < \infty$$

$$e^{-\frac{1}{2\sigma^2}(\log x - \mu)^2}; x > 0$$

$$\text{pdf of } X: f_X(x) = \frac{1}{\sqrt{2\pi}\sigma x}$$

$$\boxed{\frac{X-\mu}{\sigma}}$$

$$E(X) = \int_0^{\infty} x f(x) dx$$

$$= \int_0^{\infty} x \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{1}{2\sigma^2}(\log x - \mu)^2} dx$$

$$\frac{\log x - \mu}{\sigma} = t$$

$$\frac{1}{x} dx = \sigma dt$$

$$dx =$$

$$(\mu + \sigma t)$$

$$\frac{\sigma^2}{2}$$

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$$\left(\mu + \frac{\sigma^2}{2}\right)$$

$$E(X) = \mu + \frac{\sigma^2}{2}$$

$$Var(X) = \omega^2(\omega^2 - 1) e^{2\mu} ; \omega = \frac{\sigma^2}{2}$$

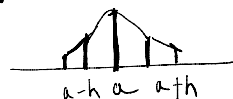
$$E(X^2) - (E(X))^2$$

Cauchy distribution:

$$f(x) = \frac{1}{\pi} \times \frac{1}{1^2 + (x-\mu)^2} \quad -\infty < x < \infty \quad | \rightarrow \mu \neq 0$$

It is a symmetric dist with the mode and median both equal to μ .

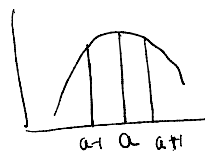
If a dist is symmetric about pt $\frac{a}{h}$
 $f(a-h) = f(a+h)$
 $h > 0$



$$f(\mu+x) = f(\mu-x)$$

~~It~~
 $E(X)$ will not exist.

$\mu=0, \lambda=1 \rightarrow$ standard Cauchy dist.



$$\frac{E|X|^r}{E|X|^r} = \int_{-\infty}^{\infty} |x|^r \frac{1}{\pi} \times \frac{1}{1 + (x-\mu)^2} dx$$

$\downarrow$ will not converge for $r \geq 1$.

Gamma distribution

$$f(x) = \frac{\lambda^p}{\Gamma(p)} e^{-\lambda x} x^{p-1}, \quad x > 0$$

$$\lambda, p > 0$$

$$\mu = E(X) = \int_0^{\infty} x f(x) dx$$

$$= \int_0^{\infty} x \frac{\lambda^p}{\Gamma(p)} e^{-\lambda x} x^{p-1} dx$$

$$\left[\frac{1}{\lambda^p \Gamma(p)} e^{-\frac{x}{\lambda}} x^{p-1} \right]$$

$\left(\lambda = \frac{1}{\lambda} \right)$

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$$\begin{aligned}
 &= \int_0^{\infty} x \cdot \frac{\alpha^p}{\Gamma(p)} \left(\frac{x}{\alpha} \right)^{p-1} e^{-x/\alpha} dx \\
 &= \frac{\alpha^p}{\Gamma(p)} \int_0^{\infty} x^p e^{-x/\alpha} dx \\
 &= \frac{\alpha^p}{\Gamma(p)} \int_0^{\infty} \left(\frac{t}{\alpha} \right)^p e^{-t} \frac{dt}{\alpha} \\
 &= \frac{1}{\alpha \Gamma(p)} \int_0^{\infty} t^p e^{-t} dt \\
 &= \frac{1}{\alpha \Gamma(p)} \cdot \Gamma(p+1) \\
 &= \frac{p \Gamma(p)}{\alpha \Gamma(p)} = \left(\frac{p}{\alpha} \right)
 \end{aligned}$$

$$\begin{aligned}
 \alpha x &= t \\
 dx &= \frac{dt}{\alpha} \\
 x &= \frac{t}{\alpha}
 \end{aligned}$$

$$\begin{aligned}
 \Gamma a &= \int_0^{\infty} e^{-x} x^{a-1} dx \\
 \Gamma a &= (a-1) \Gamma(a-1)
 \end{aligned}$$

$$\begin{aligned}
 E(x^n) &= \int_0^{\infty} x^n f(x) dx \\
 &= \frac{p(p+1) \dots (p+n-1)}{\alpha^n}
 \end{aligned}$$

$$r=2, E(x^2) = \frac{p(p+1)}{\alpha^2}$$

$$Var = E(x^2) - (E(x))^2$$

$$Var(x) = \frac{p}{\alpha^2}$$

Beta dist

$$f(x) = \frac{1}{B(m,n)} x^{m-1} (1-x)^{n-1}, \quad 0 < x < 1$$

$$B(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$m, n \geq 0$$

$$x^{m+1-1} (1-x)^{n-1} dx$$

$$\begin{aligned} \mu = E(X) &= \int_0^1 \frac{x^{m+1-1} (1-x)^{n-1} dx}{B(m, n)} \\ &= \frac{B(m+1, n)}{B(m, n)} \quad \left(\begin{array}{l} \sqrt{m+1+n} \\ = \sqrt{m+n} \\ (m+n) \end{array} \right) \end{aligned}$$

$$B(m, n) = \frac{\sqrt{m} \sqrt{n}}{\sqrt{m+n}} = \frac{m}{m+n}$$

$$\text{Var}(X) = \frac{mn}{(m+n)^2 (m+n+1)}$$